

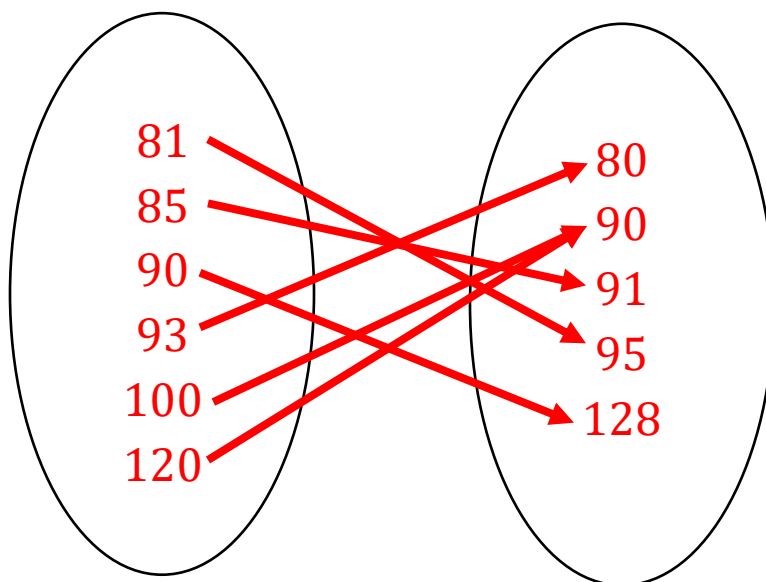
1. The table shows the maximum descent angle and maximum speed of seven roller coasters.

Roller Coaster Name	Maximum Descent Angle (degrees)	Maximum speed (miles per hour)
Kingda Ka	90	128
Millennium Force	93	80
Spiderman: Escape from Krypton	100	90
Fury 325	81	95
Orion	85	91
Top Thrill Dragster	120	90

- a. Complete the mapping diagram using the data in the table.

Maximum Descent Angle (degrees)

Maximum speed (miles per hour)



b. Complete the statements.

The relationship between the maximum descent angle and

maximum speed of the roller coasters

☒ is
☐ is not

a function. The

domain of the relationship is { 81, 85, 90, 93, 100, 120 }.

2. Tables of values for relation K and relation R are shown.

Relation K					
x	-4	-7	4	1	-2
y	6	1	17	12	7
Relation R					
x	-6	-10	2	4	0
y	-5	0	-15	-17.5	-12.5

Complete the statement.

- ☐ Only relation K could represent a
- ☒ Only relation R could represent a
- ☐ Both relation K and R could represent
- ☐ Neither relation K and R can represent a

linear relationship(s).

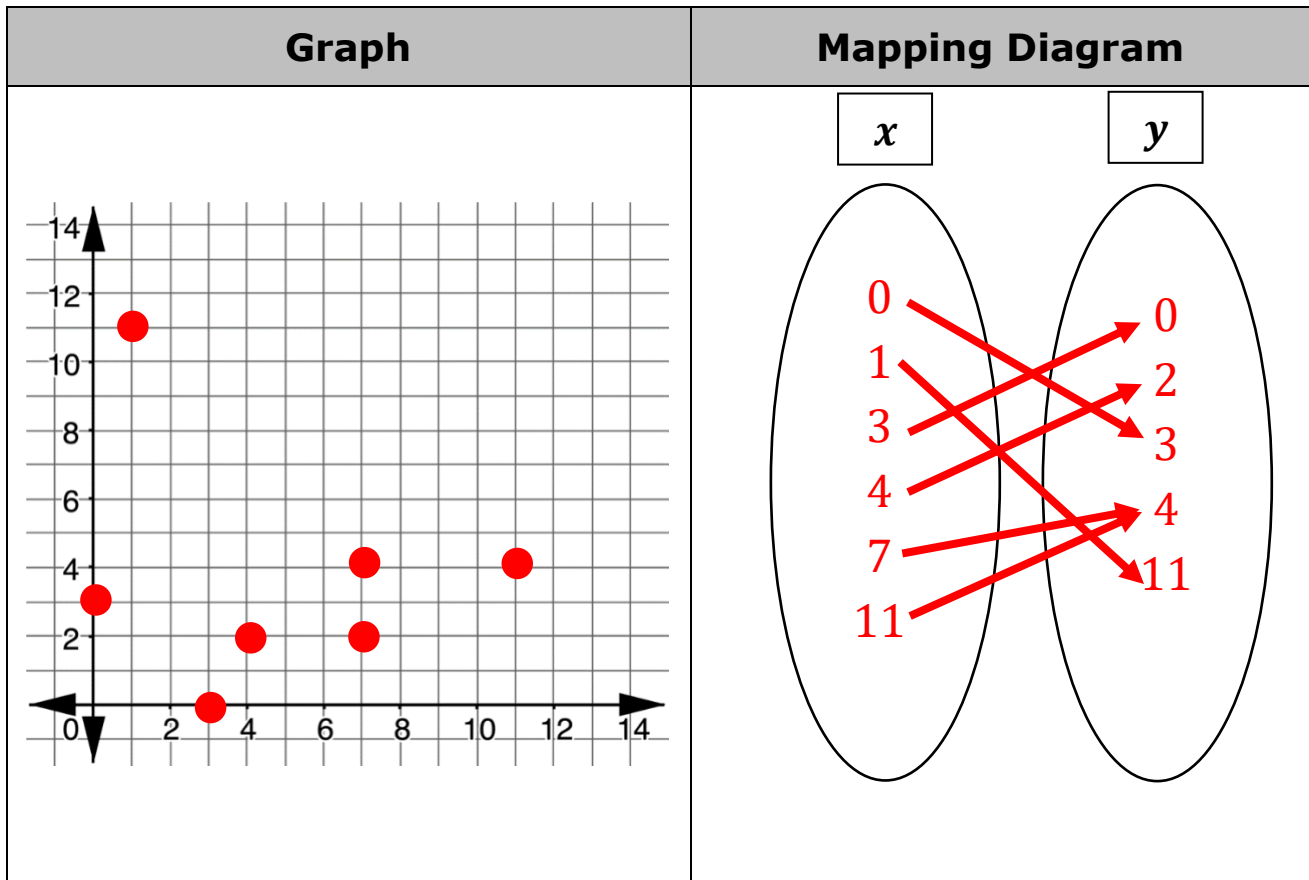
3. A relation is shown where x represents the independent variable and y represents the dependent variable.

x	11	3	7	4	1	0	7
y	4	0	2	2	11	3	4

a. Identify the domain and range of the relation. Represent the domain as a list or using an inequality.

Domain	Range
0, 1, 3, 4, 7, 11	0, 2, 3, 4, 11

- b. Complete the graph and mapping diagram to represent the same relation.



- c. Determine which ordered pair can be removed so the relation is a function.

If the ordered pair (7,2) or (7,4) is removed, then the relation is a function.

4. The table shows two groups of functions.

Group A	Group B
$y = -2(3 + x)$ $y = 5$ $y = \frac{4}{5}x^2 + 1$ $x = -y + 4$	$\frac{1}{x} = y$ $y = x^4$ $4x - 8 = 2y$ $2x + 3y = 6$

a. Complete each statement.

Group A has

- ☐ 0
- ☐ 1
- ☐ 2
- ☒ 3
- ☐ 4

equations whose graph would be a line.

Group B has

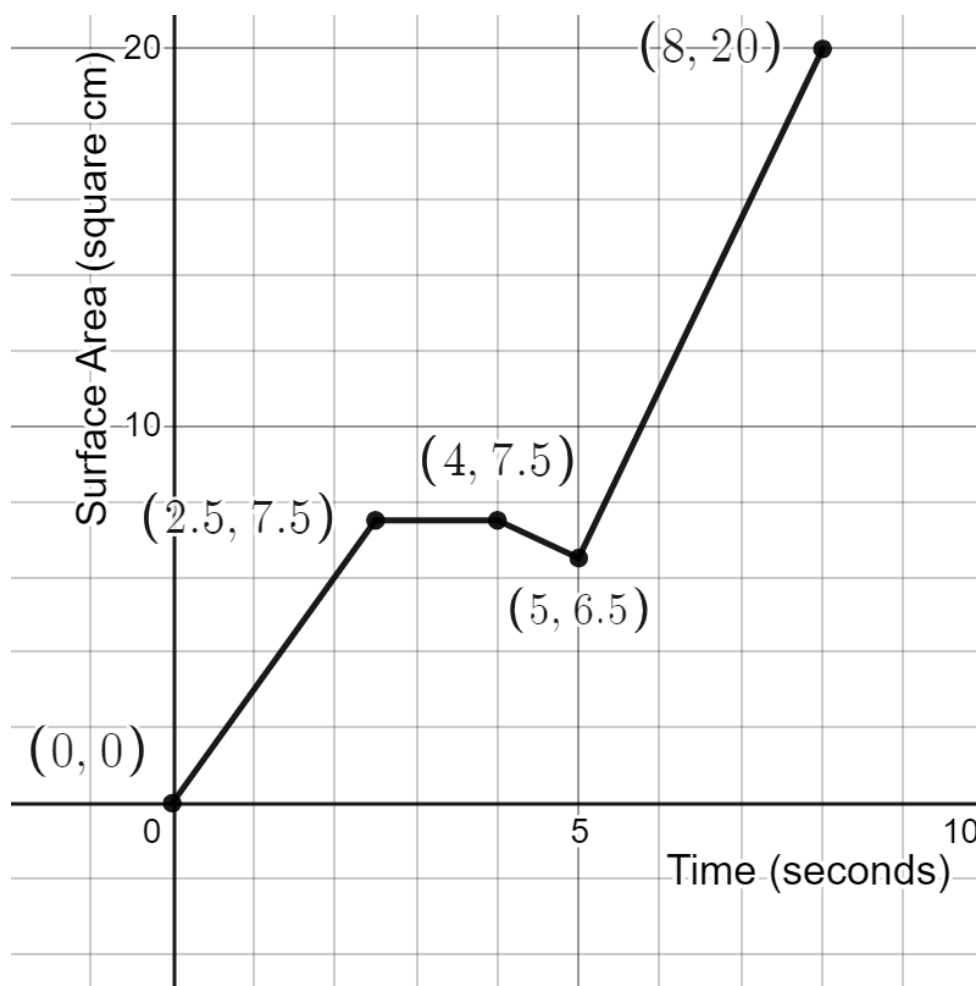
- ☐ 0
- ☐ 1
- ☒ 2
- ☐ 3
- ☐ 4

equations whose graph would be a line.

b. List the equation(s) in group A whose graph would NOT be a line.

$$y = \frac{4}{5}x^2 + 1$$

5. Jaheem was blowing up a balloon. The graph shows the surface area of the balloon, in square centimeters (cm^2), with respect to time, in seconds.



- a. What is the range of the graphical representation of Jaheem blowing up the balloon? Write your answer using inequalities.

Range	$0 \leq y \leq 20$
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- b. Which interval(s) of the domain does the graph show that the function decreases? Write your answer using inequalities.

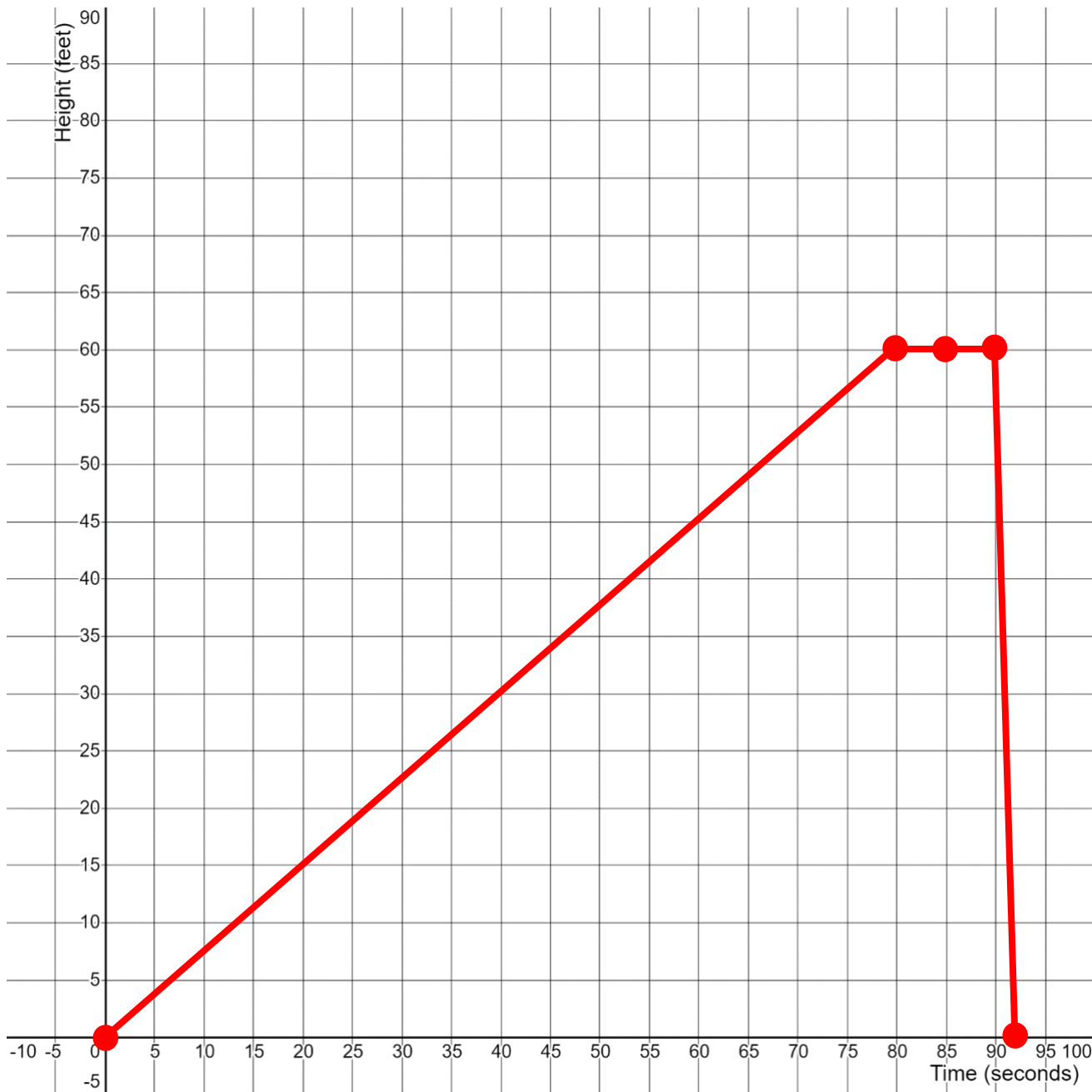
Decreasing	$4 \leq x \leq 5$
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- c. Describe what happens to the surface area of the balloon between 2.5 and 4 seconds.

The surface area remains constant.

6. Isabella is at the bottom of the stairs for a high dive with a platform at 60 feet high. It takes Isabella 80 seconds to climb to the platform. It then takes Isabella 5 seconds to walk to the edge of the platform where she pauses for another 5 seconds. Then, Isabella dives from the platform, landing in the water 2 seconds later.

a. Draw the graph of the relationship between the height of Isabella above the ground, in feet, and time, in seconds.



b. Between which times was Isabella going the fastest?

Isabella goes the fastest between 90 and 92 seconds.